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A 3-D Monte-Carlo (Potts) model for recrystallization and grain growth in polycrystalline materials

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Abstract

The evolution of microstructure during annealing of deformed polycrystalline metallic materials was modeled using a 3-D Monte-Carlo (Potts) approach in which the initial microstructure, texture (ODF), spatial distribution of the stored energy of deformation, and nucleation mechanism were carefully quantified. The modeling technique was tested using several special cases of 3-D stored-energy distributions and different textures of the as-deformed state.

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1. Introduction

Annealing is generally the final step in the thermomechanical processing of metallic materials and is critical to the optimization of their service properties. Two important processes occur during annealing: static recrystallization, driven by stored energy, and grain growth, driven by the grain-boundary (GB) surface energy. Most often, both of these processes are interrelated in time and space. Generally, the end of recrystallization is considered as the time when the last portion of the deformed material disappears. However, due to the spatial non-uniformity in stored-energy distribution, recrystallization kinetics are usually different in various locations, thus giving rise to simultaneous propagation of the recrystallization front and grain growth in already recrystallized locations.

Recrystallization is a classic subject in physical metallurgy and is of both considerable theoretical interest and industrial importance. Research and the latest developments in this subject are summarized in Refs. [1–3]. During the last decade, substantial progress in the understanding of recrystallization phenomena has been achieved by using various analytical and numerical modeling approaches [4–12]. As a result, a qualitative, and sometimes quantitative, description of microstructure and texture evolution during recrystallization has been developed [3,12,13]. Nevertheless, there appears to be no complete quantitative model which incorporates the complex microstructural features that control recrystallization kinetics, grain growth within recrystallized volumes, and corresponding texture evolution. Contemporary modeling efforts [14–17] have shown that the evolution of microstructure and texture in polycrystalline materials during annealing are closely related, and thus both must be taken into account. The problem of recrystallization and simultaneous grain growth poses an even more complicated situation.

Historically, recrystallization and grain growth have been treated separately using different techniques, e.g., cellular-automaton (CA) for recrystallization [6,8,18] and the Monte-Carlo (MC) approach for grain growth [4,17]. The MC technique faces difficulties in reproducing the linear relationship between the stored-energy gradient and the rate of migration of the recrystallization front due to so-called "model-lattice" effects. Fig. 1 illustrates the nature of the model-lattice effect which distorts the shape of the recrystallization front and associated recrystallization kinetics during standard MC simulations. Specifically, consider the recrystallization-front shape prior to an MC trial (Fig. 1a). Fig. 1b and c illustrates how two different lattice

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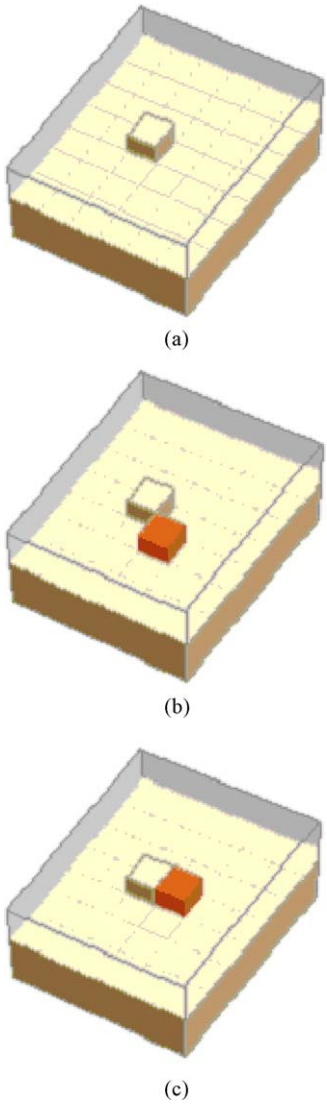

Fig. 1. Schematic illustration of the "model-lattice" effect: (a) state prior to MC orientation flip and (b and c) different post-flip states that cannot be distinguished by standard MC approaches.

points, or so-called 'model units' (MUs), can be selected within the unrecrystallized volume for the MC trial procedure. There is no difference between cases b and c in standard MC procedures; both MUs have an equal number of neighboring MUs that belong to the recrystallized (10) and to the unrecrystallized (16) volumes. It is also obvious that there is an equal change in local stored energy for configurations (a → b) and (a → c) in the case when there is no significant gradient of stored energy along the horizontal plane. However, the MUs selected for the flip trial in cases b and c have different 'coordination' relations with respect to the upper MU in configuration a. Hence, there must be a difference in the probabilities of MU reorientation for cases b and c in order to predict a realistic shape for the recrystallization front and the overall recrystallization kinetics. Ignoring this difference in standard MC approaches may result in noticeable distortions of the recrystallization front; e.g., the formation of bubbles, occlusions (as reported in Ref. [4]), "whiskers", etc. For instance, a standard MC simulation technique was employed to model recrystallization and grain growth using a 2-D modeling volume (with a triangular lattice array) in Ref. [19]. It was shown that the recrystallization front becomes unrealistically rough for high stored energies. In contrast, the CA approach does not contain probabilistic trials and, hence, is free from this particular problem. As a result, MC simulations are usually not used to simulate recrystallization. However, there has been some initial success in formulating a 2-D hybrid approach for modeling grain-growth processes running in parallel with recrystallization [20]. In addition, two relatively recent examples of the "pure" MC simulation of recrystallization and grain growth phenomena can be found in Refs. [14,21].

In Ref. [14], an approach to describe the simultaneous evolution of microstructure and texture during recrystallization was developed. For this purpose, an MC routine was proposed in which 2-D orientation and stored-energy maps are derived by electron-backscatter diffraction (EBSD), and the texture of actual materials is used as input data. Although thermally-activated grain-boundary migration in the recrystallized volumes was allowed, the interrelation between grain growth and recrystallization processes was not discussed, possibly because of the relatively small 2-D modeling volumes. In addition, the generation of new orientations during recrystallization was not considered. The nucleation of recrystallized grains was considered to be the result of recovery and was modeled as the gradual decrease of the stored energy to the same energy level as in the less distorted regions.

The problem of oriented nucleation of the recrystallized phase was addressed in detail in Ref. [21]. Here, the MC routine utilized a 3-D microstructure reconstructed from a set of in-plane and transverse EBSD sections of the as-deformed specimen. Once the 3-D digital microstructure was generated, nuclei were added to the system. In contrast to a recovery-based model for nucleation [14], the simulation routine in Ref. [21] allowed the treatment of several types of nucleation. Furthermore, the authors listed the main factors affecting texture development during recrystallization in order of their impact: (1) the orientation of the nuclei; (2) the growth rate of the new grains; (3) the location of new grains belonging to a given texture component relative one to another; and (4) the stored-energy level. It was shown that agreement with experimental observations could only be obtained by taking into account oriented nucleation, stored energy, *and* the anisotropy of GB properties. However, one of the principal assumptions of this work was that there was no local variation in stored energy within each grain and that all grains with a similar orientation contained the same stored energy. Although such an assumption represents a good first approximation, it does not correspond to detailed experimental data describing the spatial distribution of stored energy.

Motivated by recent efforts, the objective of the present work was to develop and validate a computer-modeling technique for the analysis of microstructure and texture evolution during annealing. For this purpose, an advanced model based on a 3-D Monte-Carlo (Potts) approach was developed. The model incorporated a complete description of microstructure and texture and took into account recrystallization, driven by stored energy,

and grain growth, driven by thermally-activated GB migration within the recrystallized regions.

2. Physical model of recrystallization and grain-growth phenomena

In the present work, recrystallization and grain growth were treated in the typical manner as a process consisting of nucleation and growth of new (recrystallized) grains followed by thermally-activated GB migration within the recrystallized volumes.

2.1. Nucleation

Fig. 2a shows the microstructure of the deformed material prior to recrystallization. It is commonly observed that recrystallization nuclei originate at inhomogeneities in the deformed microstructure, such as grain boundaries, shear bands, etc. Nucleation of new grains in the vicinity of prior grain boundaries, whose orientations are related to orientations of one of the parent grains, is more common at larger strains [3,12,22,23]. The average nucleation rate is given by using following phenomenological relation [3]:

$$\dot{\bar{N}} = Z(H_\Omega(t) - H_{\mathrm{MIN}}(t))V_\Omega(t)\exp\left(\frac{-Q_A}{k_B T}\right);$$
$$H_\Omega(t) - H_{\mathrm{MIN}}(t) > 0, \quad (1)$$

in which $H_\Omega(t)$ is the average stored energy of deformation, $H_{\mathrm{MIN}}(t)$ the minimum stored energy that causes the nucleation of recrystallization within the deformed material, $V_\Omega(t)$ the fraction of material for which nucleation is still possible at time t, Z a constant, Q_A the activation energy for nucleation, k_B the Boltzmann's constant, and T is the absolute temperature. In the case for which $H_\Omega(t) - \mathrm{H}_{\mathrm{MIN}}(t) \leq 0$, nucleation is probable only in localized regions with higher stored energy; hence, they are treated separately. Under isothermal conditions, the average nucleation rate is dependent only on the stored energy. The present simulation does not use time-dependent values of H_Ω and H_{MIN}. In the future, the competition between recovery and recrystallization will be modeled by introducing time-dependent values of H_Ω and H_{MIN}.

For a constant nucleation rate per unrecrystallized unit volume, the recrystallized fraction F as function of annealing time exhibits a sigmoidal behavior. This can be expressed mathematically by the well-known JMAK relation derived originally by Johnson and Mehl [24], Avrami [25], and Kolmogorov [26]. In such an instance, the dependence of F on time is described by a linear behavior in a log $(-\ln(1-F))$ versus log (t) plot.

Although it is generally accepted that oriented nucleation occurs in most recrystallization cases, many details of how nuclei of particular orientations develop remain unclear. Hence, one can use experimentally derived orientation relationships between the texture of the deformed matrix and recrystallization nuclei in modeling instead of simulating individual nucleation events. Specific orientations of the recrystallization nuclei relative to the deformed matrix have been found in many experiments (Table 1) [2,3,18,23,27–29].

The specifics of boundary motion and stored work are summarized in the sub-sections that follow.

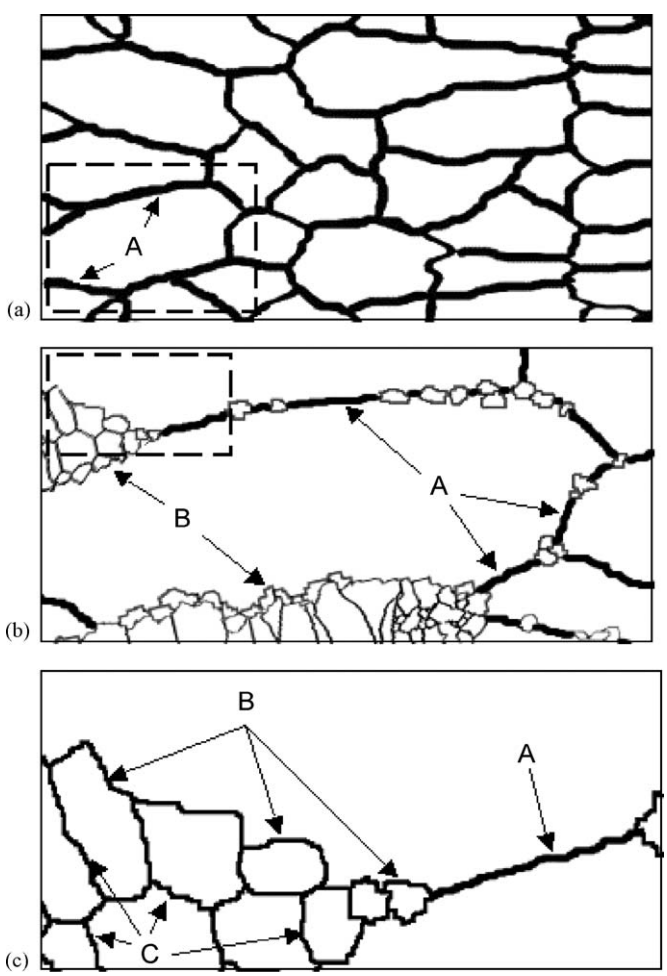

Fig. 2. Schematic illustration of microstructure: (a) prior to recrystallization, (b) selected area of (a) after recrystallization starts, and (c) selected area of (b) at higher magnification.

2.2. Grain-boundary migration

Fig. 2 presents three different types of grain boundaries considered in the modeling of recrystallization. Boundaries between the grains in the deformed material prior to recrystallization are denoted as type-A boundaries (Fig. 2a); grain boundaries between the deformed matrix and recrystallized grains as type B (Fig. 2b); and grain boundaries between recrystallized grains as type C (Fig. 2c). Type-B boundaries form the recrystallization front.

The type-A boundaries were not allowed to migrate. The curvature-driven grain-boundary velocity of type-C grain boundaries is controlled by grain-boundary energy E and mobility M as well as by the local grain-boundary curvature R [16,30]:

$$v \sim EMR^{-1}, \quad (2)$$

Table 1
Preferred orientations of recrystallized nuclei in terms of crystal lattice rotations relative to the deformed matrix orientation

Theoretical predictions [3]			Experimental measurements				
$\sum$	Angle (°)	Axis	Angle (°)	Axis	Metal	Lattice	Reference
$\sum 7$	38.21	⟨1 1 1⟩	35–45	⟨1 1 1⟩	Al	f.c.c.	[2,3,27–29]
			38	⟨1 1 1⟩	Cu	f.c.c.	[3]
			36–42	⟨1 1 1⟩	Pb	f.c.c.	[3]
$\sum 9$	38.94	⟨1 1 0⟩	35	⟨1 1 0⟩	–	b.c.c.	[12]
$\sum 13a$	22.62	⟨1 0 0⟩	23	⟨1 0 0⟩	Al	f.c.c.	[3]
			19	⟨1 0 0⟩	Cu	f.c.c.	[3]
$\sum 13b$	27.80	⟨1 1 1⟩	30	⟨1 1 1⟩	Cu	f.c.c.	[3]
			30	⟨1 1 1⟩	Ag	f.c.c.	[3]
			20–30	⟨1 1 1⟩	Nb	f.c.c.	[3]
$\sum 17a$	28.07	⟨1 0 0⟩	26–28	⟨1 0 0⟩	Pb	f.c.c.	[3]
			30	⟨1 0 0⟩	Al	f.c.c.	[3]
$\sum 19a$	26.53	⟨1 1 0⟩	25	⟨1 1 0⟩	–	b.c.c.	[2,12]
			27	⟨1 1 0⟩	Fe–Si	b.c.c.	[3]
$\sum 19b$	46.83	⟨1 1 1⟩	45	⟨1 1 1⟩	Al	f.c.c.	[19]
$\sum 21b$	44.40	⟨1 1 2⟩	20–40	⟨1 1 2⟩	Al	f.c.c.	[2,3]
$\sum 27a$	31.58	⟨1 1 0⟩	30	⟨1 1 0⟩	–	b.c.c.	[12]

The average rate of migration of the boundary between grains G_i and G_j is thus:

$$v_{G_iG_j} \sim E_{G_iG_j} M_{G_iG_j} (R_{G_i}^{-1} - R_{G_j}^{-1}), \tag{3}$$

in which R_G is the average grain-boundary curvature.

For type-B grain boundaries, the velocity of the recrystallization front moving into the unrecrystallized matrix can be described as [19]:

$$v \sim M(-ER^{-1} + H_S), \tag{4}$$

in which H_S denotes the driving force for recrystallization in terms of the stored energy of deformation.

Grain-boundary migration results in grain growth within the recrystallized material. Despite the fact that the grain-growth process is a geometrically complex phenomenon, a vast amount of experimental data has shown that a simple mathematical relation expresses the statistical behavior of a large number of grains, i.e.,

$$D^p - D_0^P = Kt \exp\left(\frac{-Q}{R_g T}\right), \tag{5}$$

in which Q is the activation energy, D and D_0 the initial and final average grain sizes, t the annealing time, R_g the gas constant, T the absolute temperature, p the grain-growth exponent, and K represents the rate constant. For so-called normal grain growth during which boundary motion is not impeded and the grain boundaries maintain a self-similar network structure, the square of the mean grain size increases linearly with time (i.e., grain-growth exponent $p = 2$) [31,32]. However, for a material with a marked initial texture and misorientation-dependent effective grain-boundary mobility, the average grain-size behavior is characterized by alternating periods of fast and slow grain growth, which can be correlated to cyclic changes in material texture [16,33].

2.3. Grain-boundary energy and mobility

The existing experimental data for the dependence of grain-boundary energy and mobility on local lattice misorientation ε are sparse [34–37]. Even with the results collected in Ref. [37], it is not possible to fully describe the grain-boundary energy and mobility as functions of exact grain-boundary misorientation. Nevertheless, from a qualitative perspective, all of the measurements are similar to those shown in Fig. 3 for boundary energy and mobility as a function of the scalar misorientation ε. In Fig. 3, the energy of grain boundaries (separating grains G_i and G_j) as a function of misorientation is assumed to follow the Read–Shockley equation [38], i.e.,

$$\begin{aligned} E_{G_iG_j} &= E_m \frac{\varepsilon_{G_iG_j}}{\varepsilon_m}\left[1 - \ln\left(\frac{\varepsilon_{G_iG_j}}{\varepsilon_m}\right)\right], && \text{for } \varepsilon_{G_iG_j} \le \varepsilon_m \\ E_{G_iG_j} &= E_m, && \text{for } \varepsilon_{G_iG_j} > \varepsilon_m \end{aligned} \tag{6}$$

in which E_m is the specific energy per unit area of a random high-angle grain boundary, and ε_m is a reference value of misorientation dividing low-angle and high-angle boundaries, usually taken as 15°.

The mobility of grain boundaries was also assumed to depend on misorientation. The following equation proposed by Rollett and Holm [39] was utilized to describe the mobility of a boundary between lattice sites S_i and S_j:

$$M_{G_iG_j} = M_m \left[1 - m \exp\left(-\frac{\varepsilon_{G_iG_j}}{10}\right)^3\right] \tag{7}$$

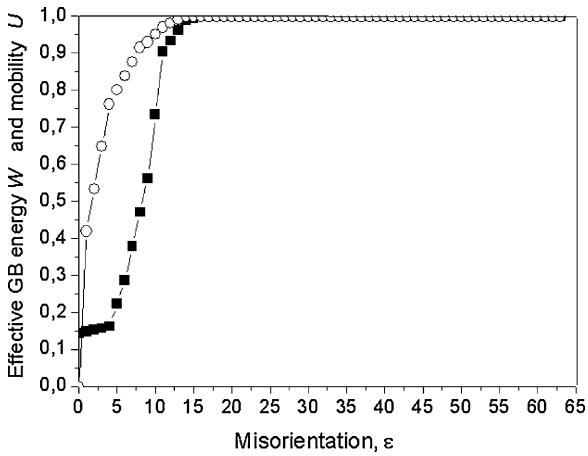

Fig. 3. Effective grain-boundary energy (○) and mobility (■) as a function of grain-boundary misorientation.

in which M_m is the maximum mobility, corresponding to random high-angle boundaries as a reference. To quantify bounds for the grain-boundary mobility, the parameter m was set equal to 0 or 1 for the isotropic case and the most anisotropic case, respectively.

For modeling purposes, normalized, instead of actual, values of energy $W(g_{G_i}, g_{G_j}) \equiv W(\varepsilon)$ and mobility $U(g_{G_i}, g_{G_j}) \equiv U(\varepsilon)$ of the grain boundary as a function of misorientation were used, viz.,

$$E = E_{\mathrm{MAX}} W(\varepsilon), \qquad M = M_{\mathrm{MAX}} U(\varepsilon), \tag{8}$$

in which E_{MAX} and M_{MAX} are the maximum possible GB energy and mobility, respectively. The values of $W(\varepsilon)$ and $U(\varepsilon)$ ($W(\varepsilon)$,$U(\varepsilon) \in [0\ldots1]$) were used in determining the elementary orientation-flip probability in present work.

2.4. Distribution of stored energy in the deformed material

As described in Eq. (4), grain-boundary motion is strongly dependent on the stored work, which is non-uniformly distributed from grain to grain. The Taylor factor is a first order approximation for the stored energy in a given grain. Most stored energy is associated with dislocations, which are distributed non-uniformly within grains. In fact, a higher level of stored energy in the vicinity of grain boundaries and triple junctions has been clearly shown by electron back-scattered diffraction analysis (EBSD) [40]. It can thus be concluded that in plastically deformed material the local stored energy associated with the model unit S often has its minimum at the center of the grain and its maximum close to the grain boundary. The dislocation density and therefore the stored energy can also exhibit intragrain maxima and minima. Local dislocation-density (and, hence, stored-energy) maxima are often associated with intersecting slip bands [3].

In previous work [12], MC simulations were conducted for the recrystallization of a material with a non-uniform stored-energy distribution in the as-deformed microstructure. However, it is often not possible to quantify the stored-energy distribution from FEM modeling, as was done in [12], or from measurements such as those based on the image quality of electron-backscatter-diffraction (EBSD) patterns. Hence, for the present work, a new analytical approach was used to describe the discrete distribution of stored energy within deformed grains for the purpose of 3-D MC simulation of recrystallization within large modeling volumes. Specifically, consider a deformed bicrystal containing a single grain boundary. For one grain in such a bicrystal (which consists of k model units (MUs) and characterized by stored energy M_T), the stored energy in model unit S_i is given by the following relation:

$$H_{S_i} = C(f(\varepsilon, L) + P_{cr}(S_i))H_{\mathrm{MAX}}; \qquad \sum_{i=1}^{k} H_{S_i} = M_T, \tag{9}$$

Here, the quantity f is a factor which depends on the misorientation ε and the distance from the grain boundary L. C is a constant; it determines the level of H at the grain boundary and the value of the stored-energy gradient close to the grain boundary. The function $P_{cr}(S_i)$ quantifies the spatial distribution of stored-work maxima within each grain. H_{MAX} is the maximum possible value of stored energy of deformation. The value of M_T can be derived experimentally by integration of the local stored-energy distribution over the grain volume as determined from EBSD data [40] or neutron diffraction or by utilizing correlations between the Taylor factor and stored energy [41]. Hence, the first term in Eq. (9) depends on the distance between the volume element and the nearby grain boundary and on the misorientation, while the second term reflects the part of dislocation-density distribution independent of the distance to the grain boundaries.

To generalize Eq. (9) to the case of a real grain having a number of grain boundaries, the stored energy per unit volume comprises the superposition of the contributions from all boundaries, i.e.,

$$H_{S_i} = C_1 \left(\frac{1}{n} \sum_{j=1}^{n} f(\varepsilon_j, L_j) + P_{cr}(S_i) \right) H_{\mathrm{MAX}};$$

$$\sum_{i=1}^{k} H_{S_i} = M_T. \tag{10}$$

Here, n denotes number of the grain boundaries which form the given grain, L_j the nearest distance to grain-boundary j in MUs, and ε_j is the misorientation of the j-GB. The desired relation between the two parts of Eq. (10) can be achieved by varying $f(\varepsilon_j, L_j)$ and $P_{cr}(S_i)$ (Fig. 4). In the simplest case, Eq. (10) can be readily modified to treat a *uniform* distribution of stored energy by omitting $P_{cr}(S_j)$ and taking $f(\varepsilon_j, L_j) = f(\varepsilon_j)$. In the more general case (Eq. (10)) with various sets of parameters can provide insight into how different stored-energy distributions affect recrystallization.

It should be kept in mind that the movement and rearrangement of dislocations due to recovery-type processes (e.g., resulting from climb and cross-slip of dislocations) can result in a significant reduction in stored energy before the initiation of recrystallization. However, the present work focused on texture and microstructure changes, rather than on the specific relationship between simulation and real time scales. Therefore, such recovery effects were not incorporated into the present modeling framework.

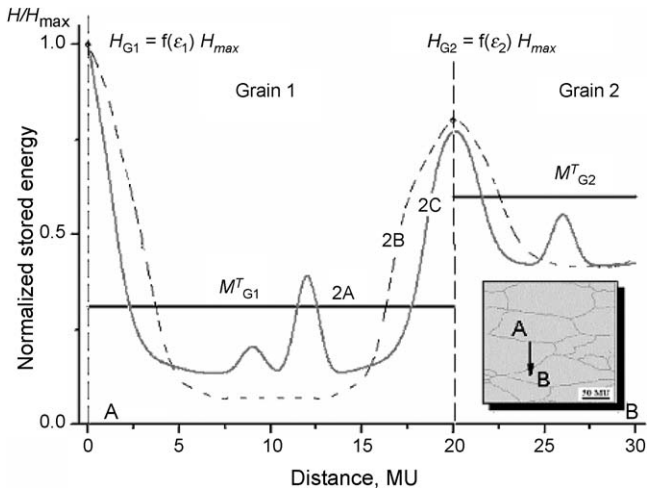

Fig. 4. Possible inhomogeneous stored-energy distributions across grains in a deformed microstructure: (M_T) homogeneous stored-energy distribution (Case 2A); (dashed) inhomogeneous stored-energy distribution for $P_{cr}(S_i) = 0$ in Eq. (9) (Case 2B); (solid line) corresponding to the case of non-zero $P_{cr}(S_i)$ (Case 2C).

3. Monte-Carlo model for recrystallization and grain growth

A distinguishing feature of the present simulations was the use of a substantially larger modeling volume, consisting of 15,625,000 MUs, than that typically employed in prior investigations. By this means, the ODF and texture of the starting (deformed) condition were reproduced precisely in the 3-D model domain.

To overcome the model-lattice effect (see Section 1 and Fig. 1), a special local shape factor, $0 \leq F(\Omega_{S_i}) \leq 1$, was introduced into the MC procedure for determining the flip probability (Fig. 5a). This factor provided a correction for the flip probabilities based on the distribution of differently oriented neighboring MUs and thus their coordination numbers. The dependence of the shape factor on the coordination number of the MU considered gives an advantage to arrangements such as (c) in Fig. 1 as compared to arrangement (b). A validation trial established that this new approach gives rise to a physically-meaningful description of the migration of the recrystallization front and helps to overcome the formation of a rough recrystallization front (Fig. 5b and c).

3.1. Initial texture and grain structure for MC simulations

The MC model domain was formed by a 3-D cubic array of MUs, each of which represented a point in a cubic lattice. Microstructure and texture evolution within the model domain was simulated as in previous work utilizing the MC approach to describe the interaction of grain growth and texture evolution [16,17]. In all calculations, the length of the side of the model domain was set equal to 250 MUs; hence, the model domain was large enough to obtain an adequate statistical representation of microstructure and texture.

For the most general 3-D case, eight parameters are needed to unambiguously define a grain boundary; namely, three terms for the orientation relationship, two parameters for the spatial orientation of the grain boundary itself, and the three components of the translation vector that characterize the displacement of

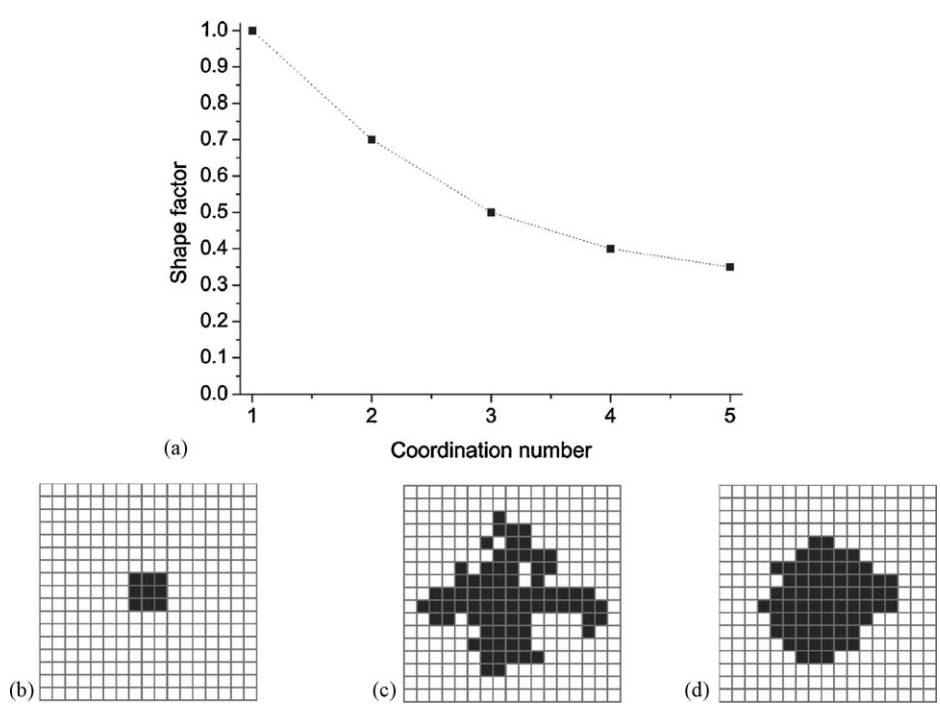

Fig. 5. (a) "Shape factor" as a function of MU coordination number. Values were obtained from MC simulations to form a smooth recrystallization front. (b–d) MC-predicted growth of an isolated recrystallization nucleus (b) (30 MU^3) within an unrecrystallized matrix after 15 MCS: (c) without using a "shape factor" and (d) using a "shape factor".

grains with respect to each other. To define the grain orientation with respect to the specimen (or model-domain) orientation, three Euler angles (φ_1, Φ, and φ_2 in Bunge notation [42,43]) were used. Specific values of the stored energy ΔH and crystallographic orientation $g = \{\varphi_1, \Phi, \varphi_2\}$ ($0 \le \varphi_1 < 360$, $0 \le \Phi < 90$, $0 \le \varphi_2 < 90$) were associated with each MU. The grain-boundary position was associated with the space between two sites having unlike orientations. Each grain was characterized by a volume equal to the number of MUs that comprised it.

A special procedure was developed to create the initial textural state in the model domain and thus to replicate the specified orientation-distribution function (ODF) as closely as possible. This procedure was based on the following probabilistic function:

$$P(\Omega_j) = \frac{1}{V_{MD}} \frac{1}{N_g(\Omega_j)} \sum_{MD} V(G_i)\chi(g(G_i)) \qquad \chi(g(G_i)) = \begin{cases} 1 & g(G_i) \in \Omega_j \\ 0 & g(G_i) \notin \Omega_j \end{cases}, \tag{11}$$

in which $g(G_i)$ is the orientation of the ith MU, $V(G_i)$ the number of MUs oriented as $g(G_i)$, N_g the total number of orientations g within the volume Ω_j, and V_{MD} is the volume of the overall model domain. From tests on model textures, it was determined that this procedure did not work well when the number of grains in the initial microstructure was less then 50, but it gave very good results when it is was equal to 100 or more.

A stochastic procedure was developed to create the initial microstructure in the model domain and to replicate experimentally-observed microstructures with specific deformed-grain aspect ratios. For this purpose, the number of grains within the model domain was determined first from the average grain size. Then, a corresponding number of MU locations was selected at random at which the grains were to be created in the model volume. The remaining MUs were then assigned to these initial grains in a probabilistic manner; i.e., a randomly selected MU site, if not yet a part of grain but adjacent to a specific grain, was assigned to this grain with a probability proportional to the prescribed aspect ratio. Such an approach enabled 99% of the model domain to be filled with grains. The remaining 1% of the MUs was then randomly assigned to adjacent grains. Then stored energies were determined according to Eq. (10) for each MU.

3.2. Grain-boundary misorientation in the MC simulations

The single parameter consisting of the lattice misorientation between neighboring grains was used to define the relative grain-boundary energy and mobility (Eqs. (6) and (7)). The present modeling routine used the (approximate) scalar misorientation ε between grains having orientations g_1 and g_2 [44], i.e.,

$$\varepsilon = \arccos\left(\frac{m_{11} + m_{22} + m_{33} - 1}{2}\right), \tag{12}$$

in which the elements m_{ij} form the matrix:

$$\mathbf{M} = \mathbf{M}_{g_1} \times \mathbf{M}_{g_2}^{-1} = \begin{pmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \\ m_{31} & m_{32} & m_{33} \end{pmatrix}, \tag{13}$$

and the rotation matrices $\mathbf{M}_{g_1}, \mathbf{M}_{g_2}$ correspond to orientations g_1 and g_2 [45,46]. If the lattice symmetry is higher than triclinic, one also needs to take into account the presence of equivalent rotations. For example, for bcc crystal structures, it is the smallest ε value from all possible pairs $\varepsilon((\mathbf{M}_{g_1})_p; (\mathbf{M}_{g_2})_q)$, where p, $q \in [1\ldots24]$ [45].

Unfortunately, the evaluation of the scalar misorientation would require much time if done for each MC trial, even using effective calculation schemes such as that based on quaternions [47]. Therefore, a database of ε as a discrete function of g_1 and g_2 was calculated and loaded into the computer memory prior to modeling. Thus, only a single search operation was performed for each MC trial.

3.3. Grain-boundary migration in the MC model

To simulate the kinetics of boundary motion with the MC technique, an MU site was selected at random. Each unit was characterized by its set of Euler angles and H value. If $H > 0$, the MU was assumed to belong to the unrecrystallized volume. A possible new MU orientation was then randomly selected from the orientations of the adjacent grains (i.e., adjacent MUs with orientations that differ from the orientation of the current MU). If the MU belonged to the recrystallization front and was already denoted as recrystallized, the new-orientation selection was restricted to orientations of the adjacent recrystallized grains. This was followed by a re-orientation attempt according to a typical MC trial procedure [16,17]. The local grain-boundary velocity is thus fully determined by the local conditions. Hence, the energy minimization for the entire model domain is equivalent to the energy minimization in the local area Ω_{S_i} close to the MU:

$$E_L = \frac{1}{2}\sum_{i=1}^{\Omega_l} W(\varepsilon_{S_iS_j})(1 - \delta_{S_iS_j}) + H_{S_i}, \tag{14}$$

in which Ω_l denotes the number of MUs in the local area around the S_i model unit, and H_{S_i} is the stored energy associated with the model unit S_i.

The probability of re-orientation of the S_i model unit from state g_k to state g_l is:

$$P = \begin{cases} U(\varepsilon)W(\varepsilon)\exp\left(\dfrac{-\Delta E_L}{k_b T_l}\right), & \Delta E_L > 0, \\ U(\varepsilon)W(\varepsilon), & \Delta E_L \le 0; \end{cases} \qquad \Delta E_L = E_L(g_l) - E_L(g_k); \qquad \varepsilon = \varepsilon(g_k, g_l). \tag{15}$$

or, when an MU belongs to the recrystallization front and not yet denoted as recrystallized:

$$P = U(\varepsilon)\frac{H(S_i)}{H_{\mathrm{MAX}}} F(\Omega_{S_i}) \exp\left(\frac{-\Delta E_L}{k_b T_l}\right), \tag{16}$$

in which $F(\Omega_{S_i})$ is the local shape factor for the MUs that belong to the recrystallization front.

The grain-boundary energy in most cases is 2–3 orders of magnitude *less* then the average stored energy H [37], and thus neglected in Eq. (16). As mentioned above, the factor $F(\Omega_{S_i})$ represents the effect of the difference in coordination number of neighboring MUs on the flip probability for the current MC trial. The quantity T_l in Eqs. (15) and (16) denotes the "model-lattice temperature", which is frequently introduced into MC simulations to overcome lattice-pinning effects.

4. MC model results for recrystallization of textured polycrystalline material

Several modeling examples were run to demonstrate the usefulness of the proposed MC-modeling approach.

The purpose of the Case 1 was to illustrate that the specifics of the evolution of recrystallization texture are mainly determined by the material texture in the starting (deformed) condition and the preferred orientations of recrystallization nuclei. Case 2 was performed to investigate in what way changes in spatial distribution of the stored energy of deformation affect simulated recrystallization kinetics. Case 3 was run to provide insight into the effect of the properties of special grain boundaries on recrystallization kinetics. Finally, two comparative simulations for Case 4 were chosen to illustrate the effect of nuclei-orientation selection on the texture formed in recrystallized material and to illustrate how the combined simulation of recrystallization and grain growth can be useful in predicting the details of final microstructure.

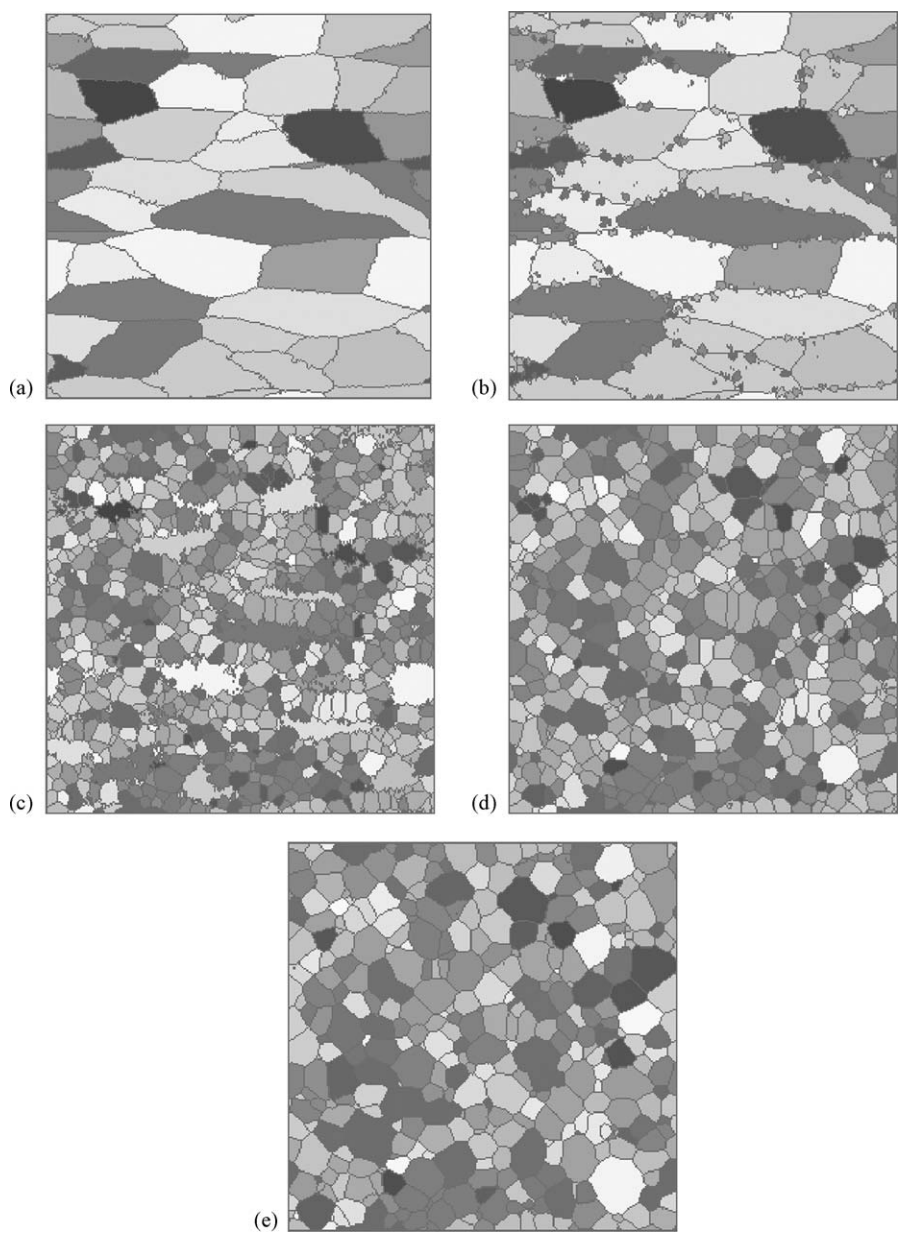

Fig. 6. MC-predicted microstructure evolution for Case 1 (cross-section #125 of 250): (a) initial structure and microstructure after (b) 5 MCS, (c) 25 MCS, (d) 50 MCS, and (e) 100 MCS.

For Cases 1, 3, and 4, the initial textures were assumed to be similar to those of as-deformed engineering beta-titanium alloys which have a bcc crystal structure. A strong one-component texture was assumed for Case 2.

In all cases, continuous nucleation of the recrystallized grains was assumed. The recrystallization nuclei were assumed to have a special boundary of $\sum 9$, $\sum 19$, or $\sum 27$ type with one of the adjacent grains, or nuclei orientations chosen at random with different probabilities specified for each particular case.

The MC simulation cases were conducted for a material with anisotropic grain-boundary properties, i.e., misorientation-dependent grain-boundary energy and mobility or, for Case 3, misorientation-dependent effective GB mobility. The recrystallization nuclei were assumed to be equiaxed (i.e., have an aspect ratio of 1:1:1) with an average volume of 15 MU^3.

Experimental observations for partially recrystallized material have shown that new (recrystallized) grains are not randomly distributed. Instead, they are often heterogeneously located in groups along grain boundaries, triple lines, or deformation bands [48]. Hence, for the present simulations it was assumed that grain boundaries (and local stored-energy maxima, Case 2C) were preferred sites for the nucleation of the recrystallized grains.

4.1. Case 1—Texture evolution during recrystallization with oriented nucleation

The purpose of Case 1 was to quantify texture evolution resulting from recrystallization with oriented nucleation. The initial microstructure was assumed to comprise grains with a volume of 8000 MU^3 and an aspect ratio of 3:3:1, to emulate the deformed microstructure of commercial titanium alloys (Fig. 6a). The initial texture within the model domain was designed to replicate the common texture of cold-drawn Timetal LCB (Ti–4.5Fe–6.8Mo–1.5Al) (Fig. 7a). The initial distribution of stored energy within each grain (i.e., the parameters in Eq. (10)) was specified with maximum values between 0.7 and $1.0 \times H_{MAX}$ at the grain boundaries (dashed line in Fig. 4). The recrystallized grains nucleated in such a way to produce a $\sum 9$

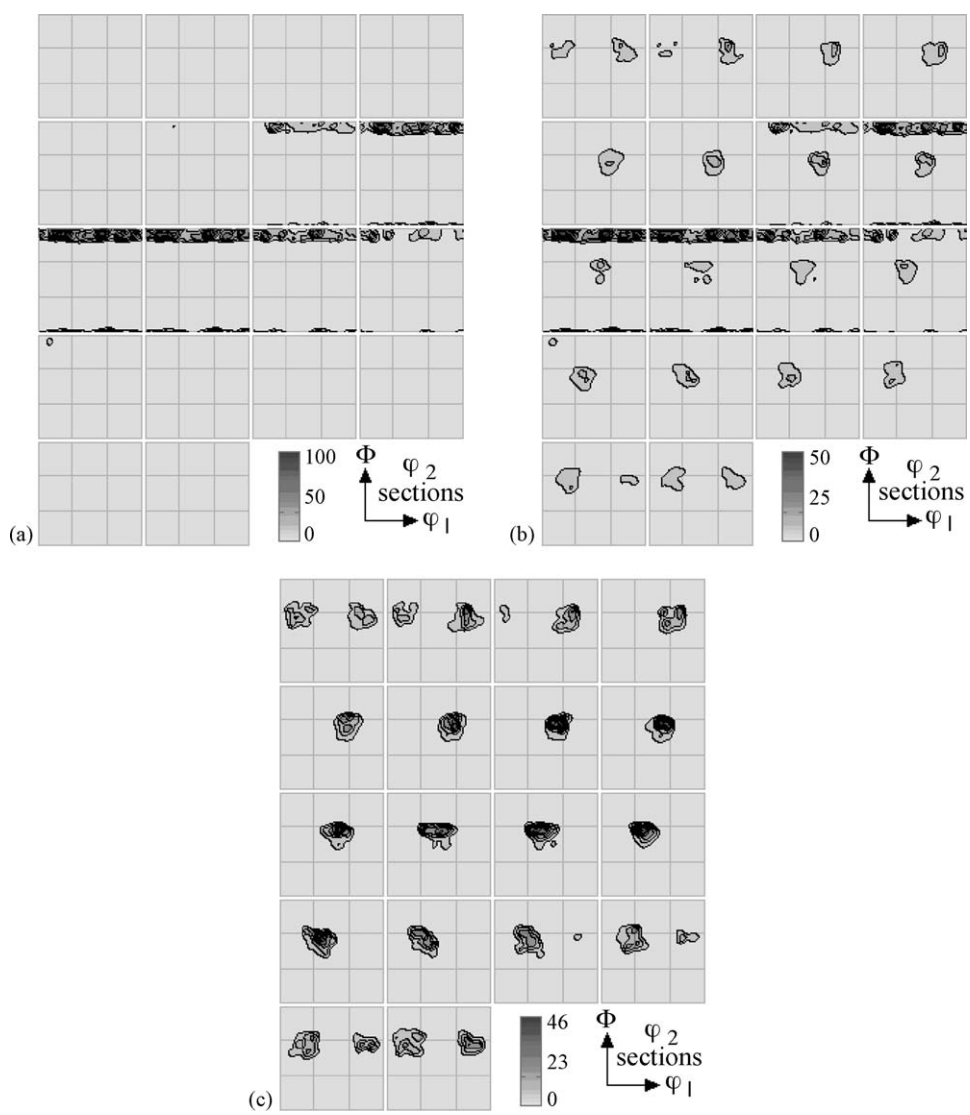

Fig. 7. MC predictions of texture evolution for Case 1: (a) initial texture and textures after (b) 15 MCS and (c) 100 MCS. Texture intensity presented in terms of a x-random scale.

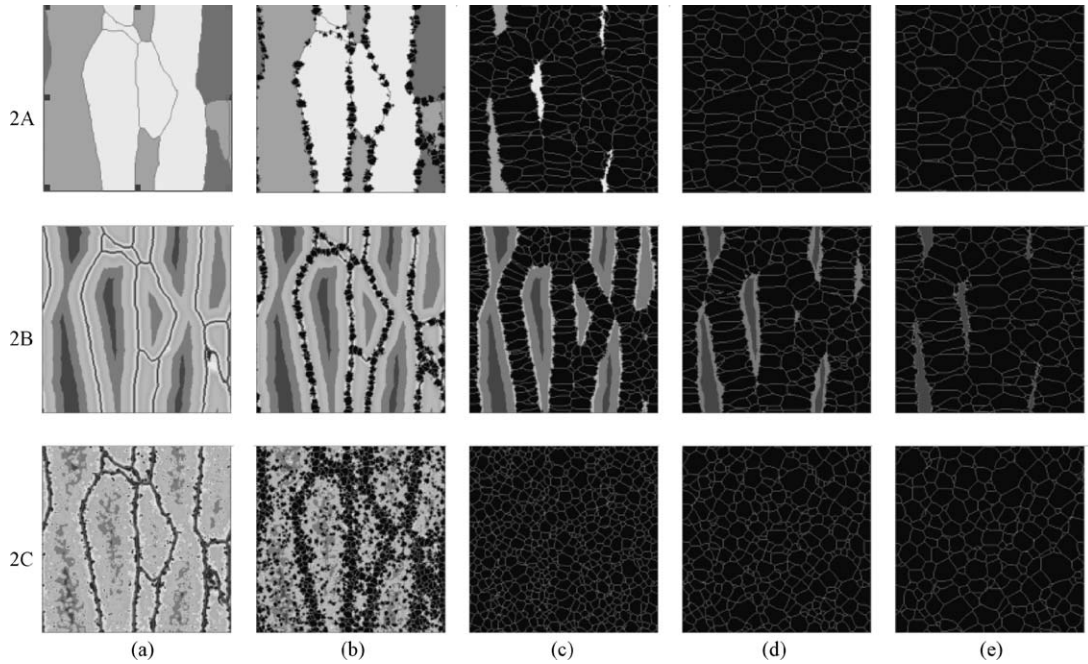

Fig. 8. MC-predicted microstructure evolution for Cases 2A, 2B, and 2C (cross-section #125 of 250): (a) initial structure and microstructure after (b) 10 MCS, (c) 50 MCS, (d) 100 MCS, and (e) 150 MCS. Stored-energy density is given via grayscale; recrystallized material is black.

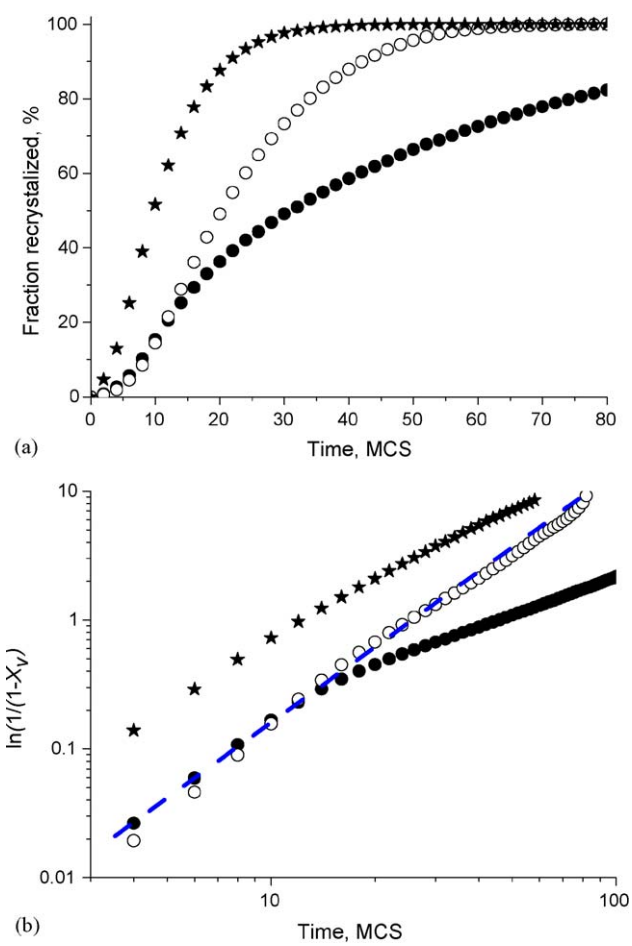

Fig. 9. (a) Comparison of MC predicted recrystallization kinetics for Cases 2A (open symbols), 2B (closed symbols), and 2C (star symbols) and (b) a fit based on the JMAK relation (line).

boundary with one of adjacent grains (probability = 0.8) or random boundary (probability = 0.2). The nucleation rate for each MCS (within the material fraction in which nucleation was possible) was assumed to be constant and equal to 10^{-4} nuclei/MU^3 per MCS. Fig. 6 summarizes MC predictions of microstructure evolution for Case 1; different grain orientations are shown by different grayscale levels. From a qualitative standpoint, the inhomogeneous nature of the microstructure developed by the end of recrystallization is indicated by the clustering of grains of similar orientations (shades of gray).

Texture predictions for Case 1 are summarized in Fig. 7. At the point at which the microstructure was 75% recrystallized (Fig. 7b), new texture maxima were formed; this finding is a

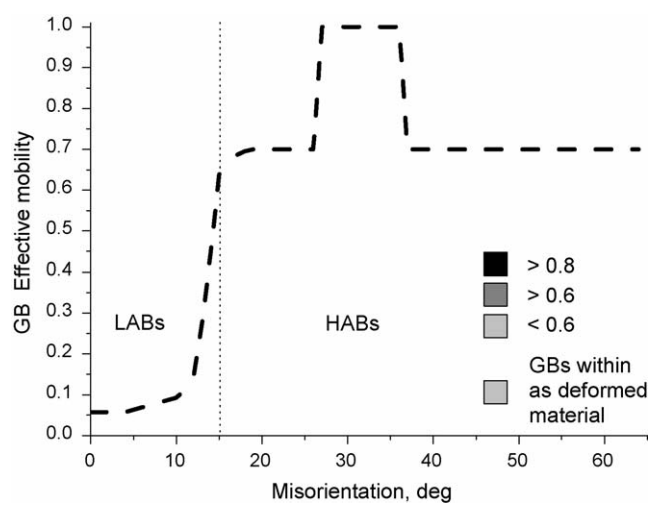

Fig. 10. Effective grain-boundary mobility as a function of misorientation for Case 3B. The grayscale levels used for GBs with different effective mobilities in the simulated microstructures (Fig. 11) are also shown.

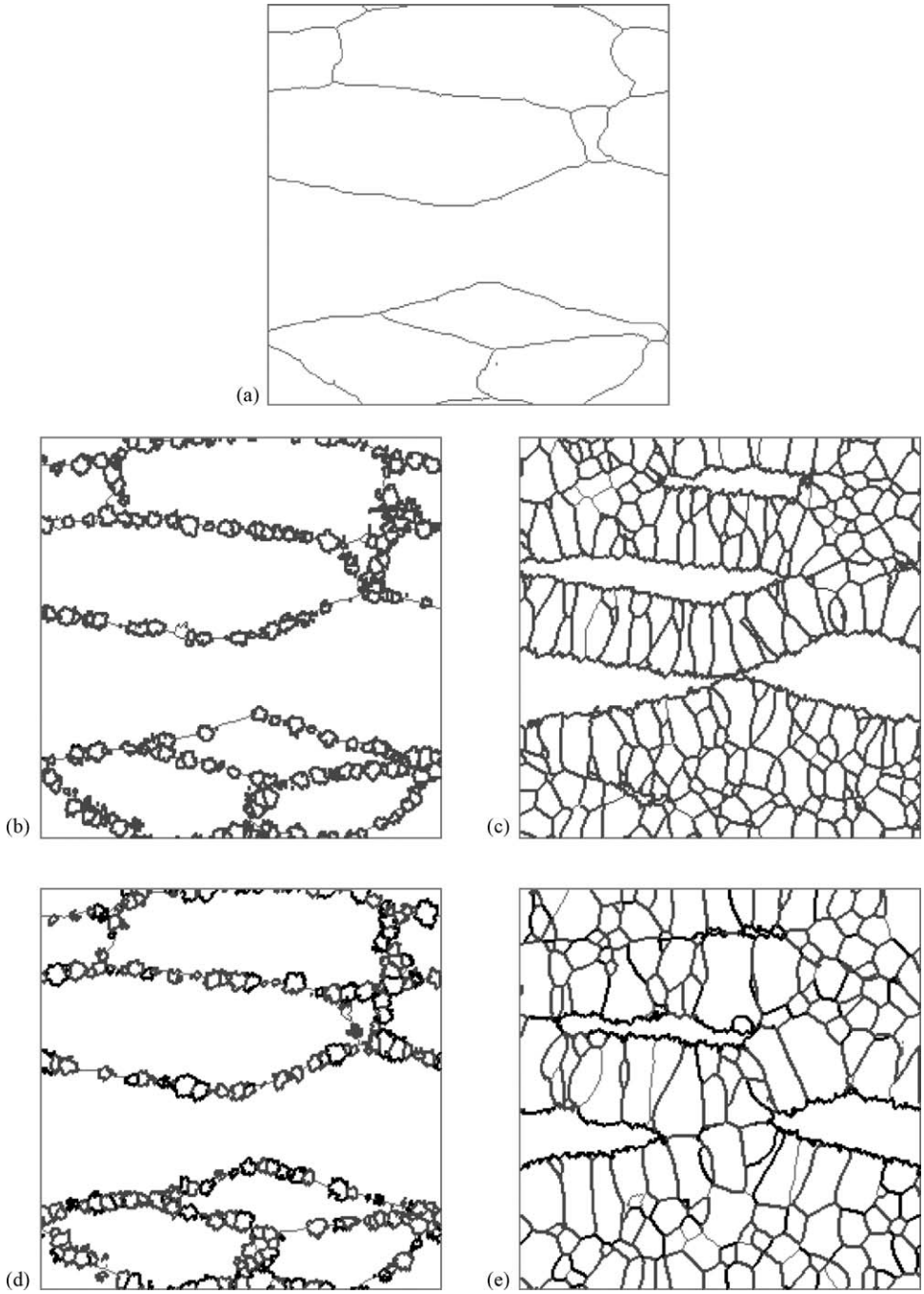

Fig. 11. MC-predicted microstructure evolution for Cases 3A (a–c) and 3B (d and e) (cross-section #125 of 250): (a) initial microstructure and microstructure after (b and d) 15 MCS and (c and e) 100 MCS. The GB grayscale level is prescribed according to the scheme in Fig. 10 to indicate the effective mobility.

result of the superposition of the initial axial texture components and the recrystallization components. As expected, initial texture maxima vanished when recrystallization was complete and strong texture components, characterized by a rotation angle of $\pm 38.9^\circ$ around the $\{1\,1\,0\}$ direction, were formed (Fig. 7c).

4.2. Case 2—Effect of inhomogeneous stored-energy distribution within the grains on recrystallization

The purpose of Case 2 was to investigate in what way an inhomogeneous spatial distribution of the stored energy of deformation affects simulated recrystallization kinetics. Initially, the deformed state was considered to be strongly textured, and recrystallization nuclei were assumed to be randomly oriented. The microstructure was assumed to be coarse (Fig. 8a); hence, the grain cross-sections were large enough to enable the modeling of situations with different stored-energy distributions. Three comparative simulations were run for Case 2. The stored-energy distribution for Case 2A was taken to be uniform within each grain (bold horizontal lines in Fig. 4, grayscale in Fig. 8a). The stored-energy distribution for Case 2B was derived using Eq. (10), with maximum values between 0.7 and $1.0 \times H_{\mathrm{MAX}}$ at the

grain boundaries (line plus symbols in Fig. 4). For Case 2C, a number of randomly-distributed stored-energy maxima were introduced within the grains (bold curve in Fig. 4). The total stored energy was equal for all three modeling cases. The nucleation rate was proportional to H_{S_i}.

Fig. 8 summarizes MC predictions of microstructure evolution for Cases 2A–C; different stored-energy densities are indicated by the different grayscale levels. From a qualitative standpoint, the most rapid recrystallization rates were found for Case 2C.

The recrystallization kinetics for Cases 2A, 2B, and 2C are quantified in more detail in Fig. 9. For the case of the inhomogeneous stored-energy distribution (Case 2B), nuclei grow quickly during the initial stage of recrystallization, namely within 0–12 MCS time interval, due to the higher level of stored-energy density close to grain boundaries. At later stages, lower recrystallization rates and noticeably different Avrami exponent (Fig. 9b) were predicted for Case 2B. For Case 2C, recrystallization was predicted to complete much faster in comparison with both Cases 2A and 2B due to a large number of additional nucleation sites distributed within the grains.

4.3. Case 3—Effect of the properties of special grain boundaries on recrystallization

Case 3 was run to provide insight into the effect of the properties of special grain boundaries on recrystallization kinetics. For the baseline Case 3A, all high-angle grain boundaries were assumed to have a similar effective mobility value ($M = 0.7$). For Case 3B, the effective grain-boundary mobility for special grain boundaries ($\sum 7$, $\sum 9$, and $\sum 19$) was assumed to be ~40% higher than that for arbitrary high-angle boundaries. Most of these special boundaries have a scalar misorientation between 25° and 37°. Hence, an effective mobility as shown in Fig. 10 was assumed for Case 3B. Nevertheless, the precise description of the properties of special boundaries is possible only by using the effective mobility as function of the exact misorientation. In practice, this function has been estimated only for some points and rotations about high symmetry lattice directions, but not for all misorientation space. Therefore, the proposed approach accounted for special boundaries in an average sense only. The nucleation rate was 10^{-4} nuclei/MU^3 per MCS.

Cases 3A and 3B used the initial texture commonly found in conventionally hot-rolled sheet of VT22 (Ti–5Al–5Mo–5V–1Cr–1Fe). In this instance, nuclei were able to form special grain boundaries of $\sum 9$, $\sum 19$, or $\sum 27$ type with one of the adjacent grains or random boundaries in order to approximate those found previously for the recrystallization of bcc metallic materials (Table 1). The probabilities used to determine the nucleation conditions were $P(\sum 9) = 0.4$, $P(\sum 19) = 0.3$, $P(\sum 27) = 0.3$, and $P(\text{random}) = 0$. The initial grain size was assumed to be coarse (Fig. 11a) in order to enable the easy visualization of the effective grain-boundary mobility during recrystallization (Fig. 11b–e).

Fig. 11 summarizes predictions of microstructure evolution in the RD–ND plane of VT22 sheet for Case 3. Due to the special orientation relationship between matrix and recrystallized material, it is clear that the effective mobility of the recrystallization front in Case 3B is higher due to the low fraction of the intermediate and low mobility grain boundaries at the recrystallization front.

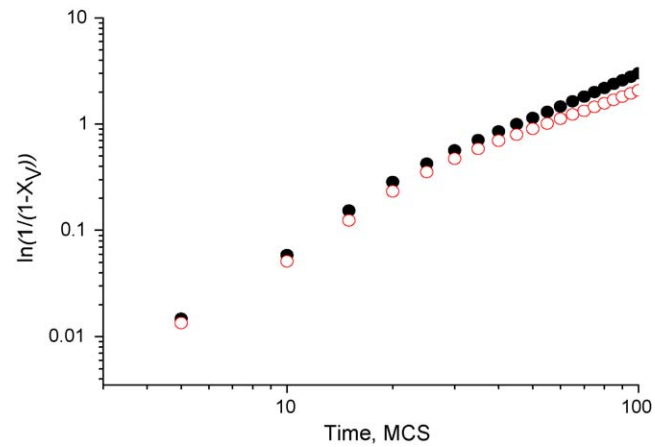

Fig. 12. MC-predicted recrystallization kinetics for Cases 3A (open symbols) and 3B (closed symbols).

The predicted recrystallization kinetics for Cases 3A and 3B are compared in Fig. 12. The kinetics for Case 3A were slower than those for Case 3B, which provides a slightly better (linear) fit to classical JMAK behavior. As is evident from Fig. 11e, only high mobility elements of the recrystallization front survive at the intermediate stages of recrystallization. Thus, the retardation of recrystallization, typical in numerical simulations of recrystallization [49], was not very pronounced in Case 3B.

4.4. Case 4—Evolution of texture/microstructure during recrystallization and grain growth

The purpose of Case 4 was to illustrate how the combined simulation of recrystallization and grain growth can be useful in predicting the details of final microstructure. The effect of nuclei-orientation selection on the texture formed in recrystallized material was also investigated. Cases 4A and 4B used the initial texture and microstructure commonly found in conventionally hot-rolled sheet of VT22 alloy. Hence, nuclei were allowed to form special grain boundaries of $\sum 9$, $\sum 19$, or $\sum 27$ type with one of the adjacent grains, similar to Case 3 simulations. The probabilities used to determine the nucleation conditions were $P(\sum 9) = 0.2$, $P(\sum 19) = 0.1$, $P(\sum 27) = 0.1$, and $P(\text{random}) = 0.6$ for Case 4A and $P(\sum 9) = 0.4$, $P(\sum 19) = 0.2$, $P(\sum 27) = 0.2$, and $P(\text{random}) = 0.2$ for Case 4B. The initial microstructure was the same as for Case 1, and the nucleation rate was 10^{-4} nuclei/MU^3 per MCS.

Microstructure-evolution predictions for Cases 4A and 4B are shown in Figs. 13 and 14, respectively. Furthermore, there was some small evidence of the formation of spatially inhomogeneous textures. Specifically, Fig. 15 summarizes the predicted ODFs after the completion of recrystallization for the simulated microstructures of Cases 4A and 4B. The resulting textures had

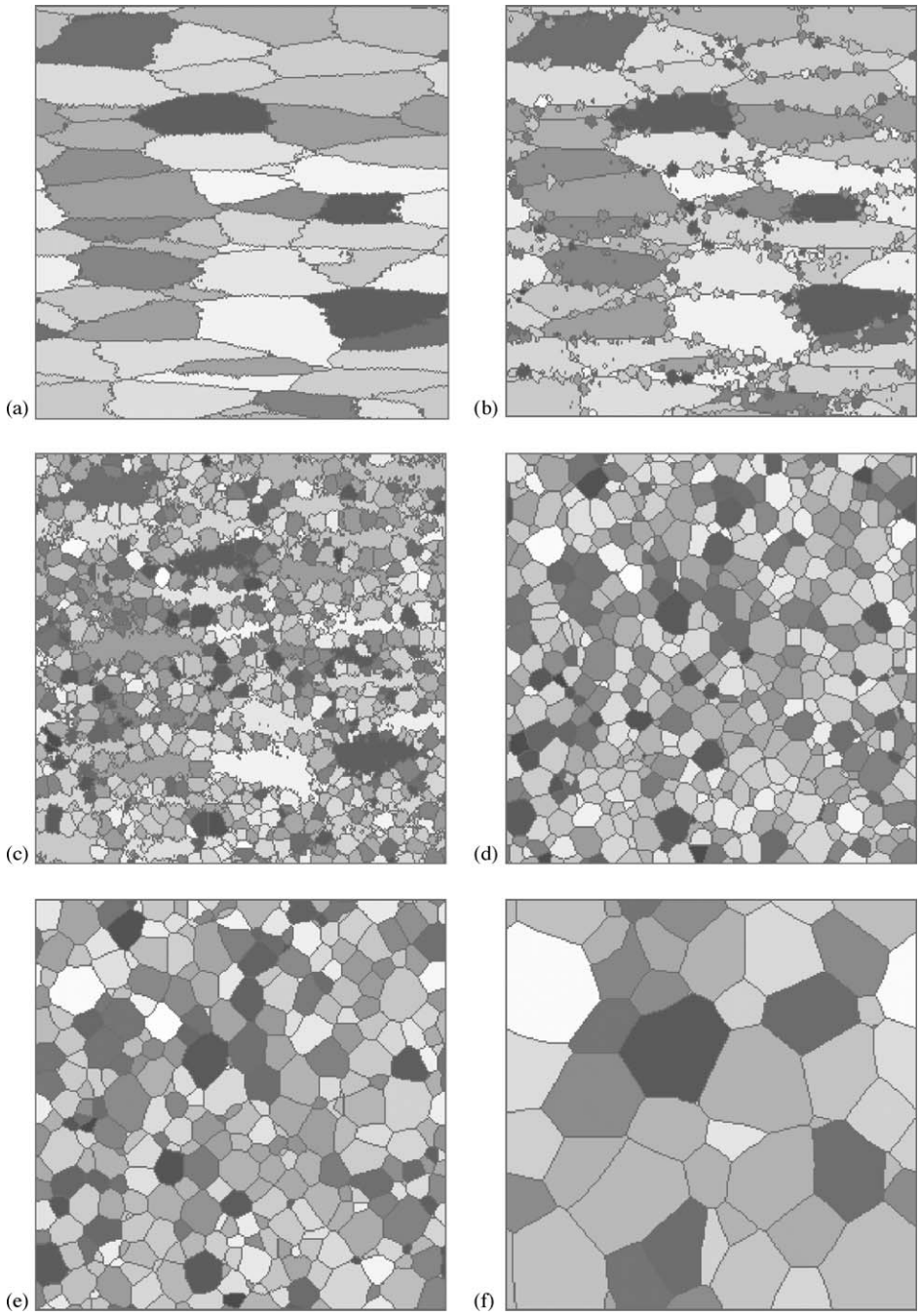

Fig. 13. MC-predicted microstructure evolution for Case 4A (cross-section #125 of 250): (a) initial microstructure and microstructure after (b) 5 MCS, (c) 15 MCS, (d) 50 MCS, (e) 100 MCS, and (f) 1000 MCS.

maxima at similar angular positions. However, the intensity of the texture was found to be different. Stronger texture formation was predicted for Case 4B due to the smaller number of random nuclei.

Fig. 16 compares recrystallization kinetics for Cases 1 and 4A, 4B. The results for Cases 4A and 4B were similar and fit classical JMAK kinetics better. Due to the more-pronounced initial texture and lower probability of the formation of random nuclei in Case 1, a lower average effective mobility was predicted for the recrystallization front. Therefore, recrystallization in Case 1 was slower; there was also a tendency for additional slowing when the recrystallized fraction reached 70%.

Simulation results for Cases 4A and 4B at longer times (∼200–1000 MCS) shed light on the details of the curvature-driven grain growth that follows recrystallization. Even though both cases used the same initial texture, the textures of the recrystallized material were different due to different nucleation conditions (Fig. 15). Hence, the long-time behavior revealed the influence of texture-controlled grain growth (Fig. 17). As

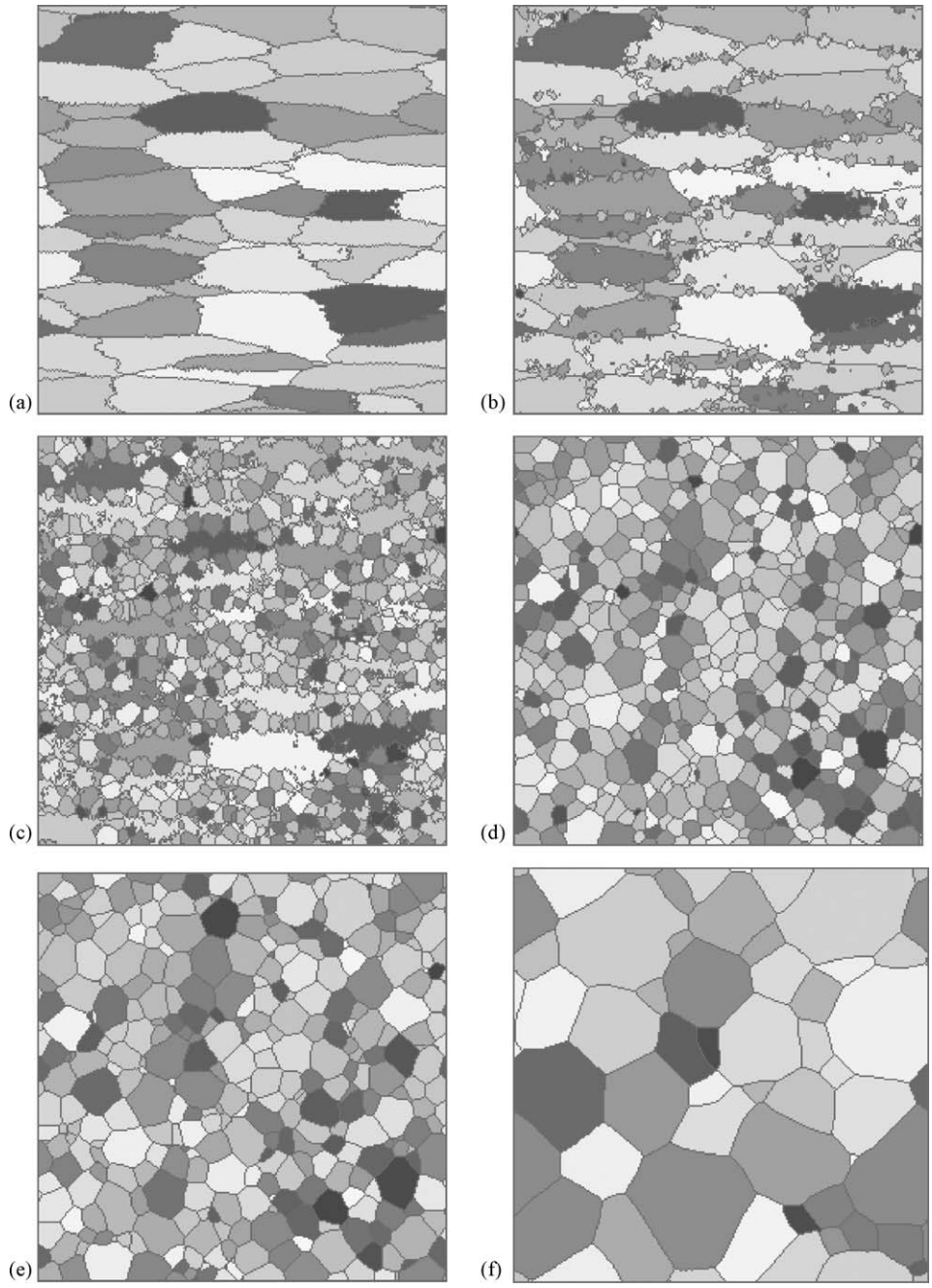

Fig. 14. MC-predicted microstructure evolution for Case 4B (cross-section #125 of 250): (a) initial microstructure and microstructure after (b) 5 MCS, (c) 15 MCS, (d) 50 MCS, (e) 100 MCS, and (f) 1000 MCS.

found in previous MC work and experimental observations [16,17,50,51], the present results demonstrated that the grain growth in less-highly textured material (Case 4A) is close to that described by Eq. (5), i.e., normal grain growth. On the other hand, material with a more pronounced recrystallization texture (Case 4B) exhibited noticeable deviations in grain growth kinetics, i.e. slower grain growth before 500 MCS and faster growth after 500 MCS. Further analysis of GB statistics from the MC simulations showed that the fraction of low-angle, low mobility GBs for Case 4B at 100 MCS (i.e., after the completion of recrystallization) was noticeably larger than for Case 4A (Fig. 18a) but became lower after 500 MCS (Fig. 18b). Therefore, the average velocity for the grain boundaries in Case 4B increased. It should also be mentioned that the size of the largest grain was 77 MU for Case 4B versus 72 MU for Case 4A after 500 MCS, while at the same time the *average* grain size for Case 4B was smaller. This difference in the size of the largest grain increased with time; the final sizes were 88 MU and 114 MU for Cases 4A and 4B, respectively.

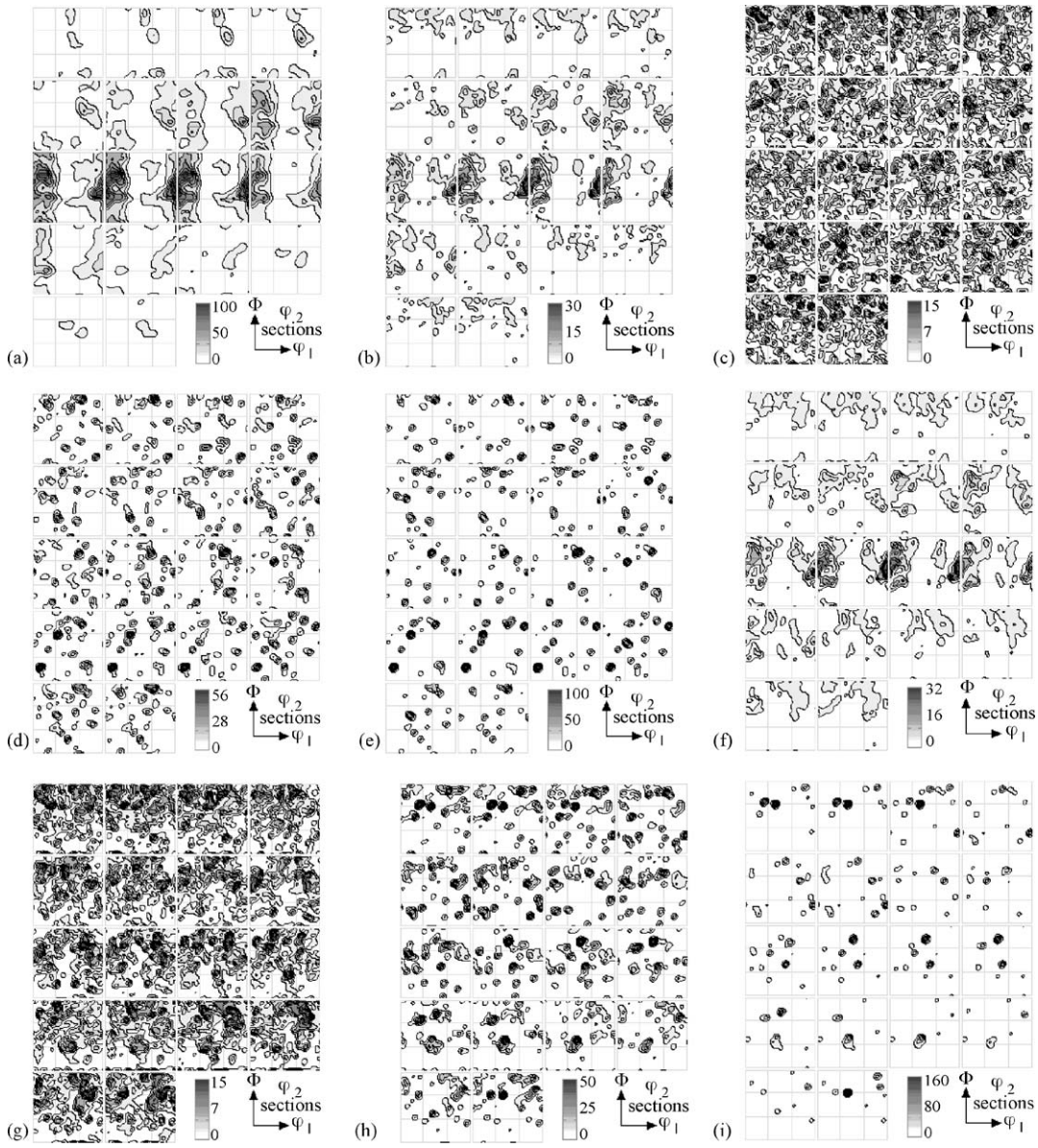

Fig. 15. MC predictions of texture evolution for Case 4: (a) initial texture, (b–e) MC-predicted texture after 15, 100, 500, and 1000 MCS, respectively, for Case 4A; and (f–i) MC-predicted texture after 15, 100, 500, and 1000 MCS, respectively, for Case 4B. Texture intensity presented in terms of a x-random scale.

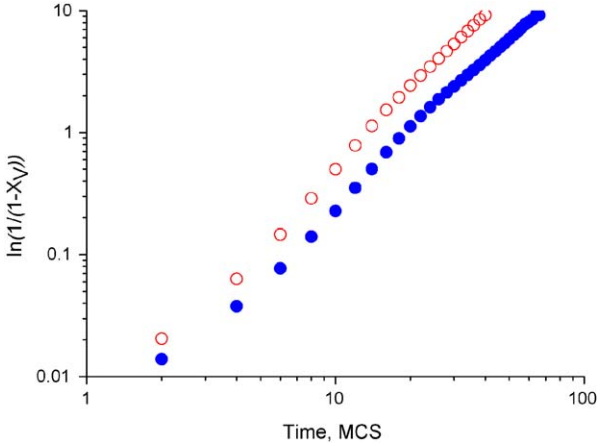

Fig. 16. Comparison of MC predicted recrystallization kinetics for Case 1 (closed symbols) and Cases 4A and 4B (open symbols).

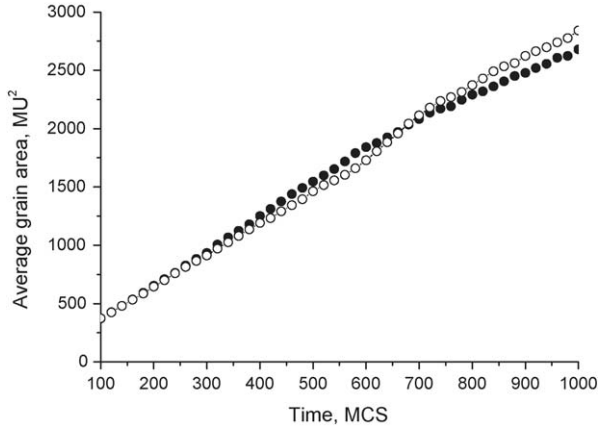

Fig. 17. MC-simulated grain-growth kinetics for Case 4A (closed symbols) and Case 4B (open symbols).

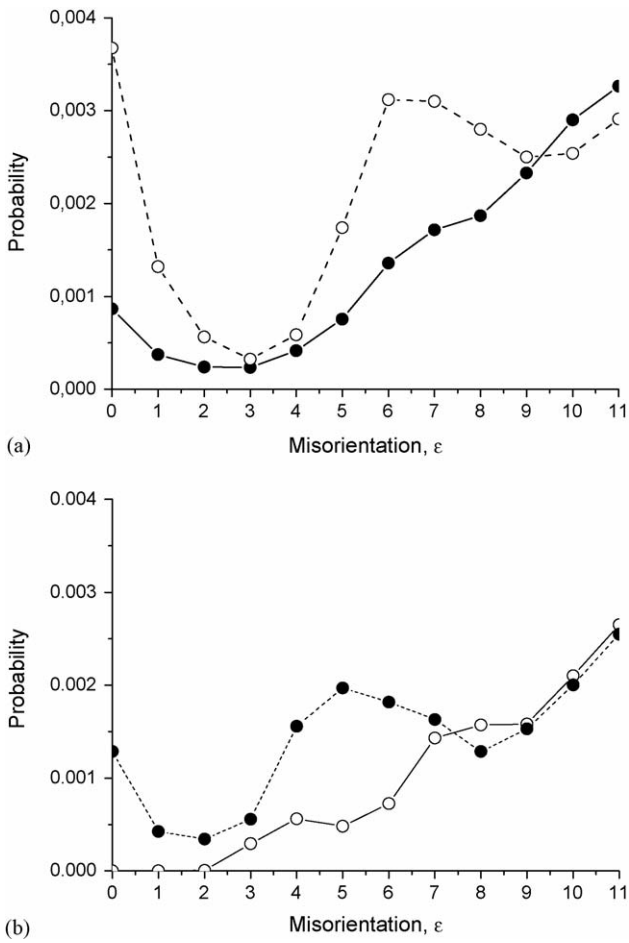

Fig. 18. Distribution of low-angle GBs for Case 4A (closed symbols) and 4B (open symbols) after: (a) 100 MCS or (b) 500 MCS.

5. Summary and conclusions

The recrystallization of polycrystalline metallic materials was modeled using a 3-D Monte-Carlo (Potts) approach in which initial microstructure, texture (ODF), spatial distribution of the stored energy of deformation, and nucleation mechanism are incorporated. Several simulations were performed to confirm the validity and usefulness of the approach, namely, the simulation of the recrystallization of textured materials when only specially-oriented nuclei are allowed to grow. It was shown that the model predictions are in good agreement with theoretical JMAK kinetics and yield realistic recrystallized microstructures. By allowing intragrain nucleation due to local stored-energy maxima, much faster recrystallization rates were obtained. It was also demonstrated that grain growth within the recrystallized volumes does not affect recrystallized-fraction kinetics, but has a significant effect on final material texture and microstructure as well as on subsequent grain-growth kinetics.

The advanced 3-D MC approach proposed in the present work provides a powerful tool to describe the effect on recrystallization of an inhomogeneous distribution of stored energy within the grains by using phenomenological parameters to emulate intragrain gradients. However, the lack of quantitative experimental data about such distributions poses an important challenge in the precise validation of the model, thus warranting future research.

Acknowledgements

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